

## OHRID ST PAUL THE APOSTLE, North Macedonia

WMO: 135780

Lat: 41.180N

Lon: 20.742E

Elev: 2313

StdP: 13.51

Time Zone: 1.00 (EUC)

Period: 96-19

WBAN: 99999

## Annual Heating, Humidification, and Ventilation Design Conditions

| Coldest Month | Heating DB |      | Humidification DP/MCDB and HR |     |      |      |      |      | Coldest Month WS/MCDB |      |      |      | MCWS/PCWD to 99.6% DB |      | WSF   |
|---------------|------------|------|-------------------------------|-----|------|------|------|------|-----------------------|------|------|------|-----------------------|------|-------|
|               |            |      | 99.6%                         |     |      | 99%  |      |      | 0.4%                  |      | 1%   |      |                       |      |       |
|               | 99.6%      | 99%  | DP                            | HR  | MCDB | DP   | HR   | MCDB | WS                    | MCDB | WS   | MCDB | MCWS                  | PCWD |       |
| (a)           | (b)        | (c)  | (d)                           | (e) | (f)  | (g)  | (h)  | (i)  | (j)                   | (k)  | (l)  | (m)  | (n)                   | (o)  | (p)   |
| 1             | 15.9       | 20.7 | 7.0                           | 8.6 | 20.7 | 11.7 | 10.9 | 24.2 | 20.7                  | 37.5 | 18.5 | 35.4 | 7.3                   | 340  | 0.459 |

## Annual Cooling, Dehumidification, and Enthalpy Design Conditions

| Hottest Month | Hottest Month DB Range | Cooling DB/MCWB |      |      |      |      |      | Evaporation WB/MCDB |      |      |      |      |      | MCWS/PCWD to 0.4% DB |      |
|---------------|------------------------|-----------------|------|------|------|------|------|---------------------|------|------|------|------|------|----------------------|------|
|               |                        | 0.4%            |      | 1%   |      | 2%   |      | 0.4%                |      | 1%   |      | 2%   |      |                      |      |
|               |                        | DB              | MCWB | DB   | MCWB | DB   | MCWB | WB                  | MCDB | WB   | MCDB | WB   | MCDB | MCWS                 | PCWD |
| (a)           | (b)                    | (c)             | (d)  | (e)  | (f)  | (g)  | (h)  | (i)                 | (j)  | (k)  | (l)  | (m)  | (n)  | (o)                  | (p)  |
| 7             | 26.0                   | 87.8            | 66.9 | 84.7 | 66.2 | 82.5 | 65.6 | 70.2                | 81.7 | 68.6 | 80.4 | 67.4 | 79.2 | 4.8                  | 200  |

| Dehumidification DP/MCDB and HR |       |      |      |      |      |      |      |      | Enthalpy/MCDB |      |      |      |      |      | Extreme<br>Max WB |
|---------------------------------|-------|------|------|------|------|------|------|------|---------------|------|------|------|------|------|-------------------|
| 0.4%                            |       |      | 1%   |      |      | 2%   |      |      | 0.4%          |      | 1%   |      | 2%   |      |                   |
| DP                              | HR    | MCDB | DP   | HR   | MCDB | DP   | HR   | MCDB | Enth          | MCDB | Enth | MCDB | Enth | MCDB |                   |
| (a)                             | (b)   | (c)  | (d)  | (e)  | (f)  | (g)  | (h)  | (i)  | (j)           | (k)  | (l)  | (m)  | (n)  | (o)  | (p)               |
| 66.3                            | 105.5 | 76.8 | 64.5 | 98.9 | 75.4 | 62.8 | 93.1 | 73.9 | 35.7          | 81.8 | 34.3 | 80.4 | 33.2 | 79.4 | 77.0              |

## Extreme Annual Design Conditions

| Extreme Annual WS |      |      | Extreme Annual Temperature |      |                    |     | n-Year Return Period Values of Extreme Temperature |     |            |     |            |     |            |      |      |     |
|-------------------|------|------|----------------------------|------|--------------------|-----|----------------------------------------------------|-----|------------|-----|------------|-----|------------|------|------|-----|
|                   |      |      | Mean                       |      | Standard Deviation |     | n=5 years                                          |     | n=10 years |     | n=20 years |     | n=50 years |      |      |     |
| 1%                | 2.5% | 5%   | Min                        | Max  | Min                | Max | Min                                                | Max | Min        | Max | Min        | Max | Min        | Max  |      |     |
| (a)               | (b)  | (c)  | (e)                        | (f)  | (g)                | (h) | (i)                                                | (j) | (k)        | (l) | (m)        | (n) | (o)        | (p)  |      |     |
| (4) 17.3          | 14.7 | 12.7 | DB                         | 10.2 | 93.0               | 4.5 | 2.4                                                | 6.9 | 94.7       | 4.2 | 96.1       | 1.7 | 97.4       | -1.6 | 99.1 | (4) |
| (5)               |      |      | WB                         | 9.2  | 73.4               | 4.2 | 1.9                                                | 6.2 | 74.8       | 3.7 | 75.9       | 1.3 | 77.0       | -1.7 | 78.4 | (5) |

## Monthly Climatic Design Conditions

|                                                                        |             | Annual | Jan   | Feb   | Mar   | Apr   | May   | Jun   | Jul   | Aug   | Sep   | Oct   | Nov   | Dec   |  |  |
|------------------------------------------------------------------------|-------------|--------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|--|--|
|                                                                        |             | (d)    | (e)   | (f)   | (g)   | (h)   | (i)   | (j)   | (k)   | (l)   | (m)   | (n)   | (o)   | (p)   |  |  |
| (6) Temperatures, Degree-Days and Degree-Hours                         | DBAvg       | 52.5   | 34.9  | 37.4  | 43.3  | 50.2  | 58.4  | 65.4  | 69.6  | 70.0  | 62.2  | 53.9  | 45.6  | 38.0  |  |  |
|                                                                        | DBStd       | 13.44  | 6.71  | 6.68  | 6.07  | 5.91  | 4.61  | 5.23  | 4.02  | 3.90  | 5.04  | 5.75  | 6.33  | 6.87  |  |  |
|                                                                        | HDD50       | 1670   | 469   | 355   | 217   | 64    | 2     | 0     | 0     | 0     | 0     | 32    | 158   | 373   |  |  |
|                                                                        | HDD65       | 4974   | 934   | 774   | 671   | 443   | 207   | 57    | 10    | 7     | 110   | 343   | 582   | 836   |  |  |
|                                                                        | CDD50       | 2585   | 0     | 1     | 11    | 71    | 263   | 463   | 608   | 619   | 367   | 154   | 26    | 2     |  |  |
|                                                                        | CDD65       | 412    | 0     | 0     | 0     | 0     | 3     | 69    | 153   | 160   | 27    | 0     | 0     | 0     |  |  |
|                                                                        | CDH74       | 4243   | 0     | 0     | 0     | 3     | 75    | 702   | 1581  | 1605  | 273   | 4     | 0     | 0     |  |  |
|                                                                        | CDH80       | 1133   | 0     | 0     | 0     | 0     | 4     | 158   | 459   | 470   | 42    | 0     | 0     | 0     |  |  |
| (14) Wind                                                              | WSAvg       | 4.9    | 5.1   | 5.6   | 5.7   | 4.8   | 4.3   | 4.4   | 4.6   | 4.5   | 4.7   | 4.6   | 4.8   | 5.6   |  |  |
| (15) Precipitation                                                     | PrecAvg     | 30.4   | 2.9   | 2.6   | 2.4   | 2.8   | 2.3   | 1.4   | 1.2   | 1.1   | 2.8   | 3.6   | 3.9   | 3.5   |  |  |
|                                                                        | PrecMax     | 45.5   | 5.7   | 6.6   | 4.1   | 4.4   | 4.4   | 3.1   | 3.6   | 3.3   | 5.8   | 6.5   | 7.4   | 7.1   |  |  |
|                                                                        | PrecMin     | 20.8   | 0.2   | 0.7   | 0.2   | 0.6   | 0.9   | 0.4   | 0.0   | 0.2   | 0.4   | 0.1   | 1.0   | 0.2   |  |  |
|                                                                        | PrecStd     | 6.6    | 1.6   | 1.4   | 1.1   | 0.9   | 1.1   | 0.7   | 1.0   | 0.7   | 1.5   | 1.7   | 1.6   | 1.9   |  |  |
| (19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures | 0.4%        | DB     | 52.6  | 57.2  | 64.9  | 73.2  | 80.1  | 88.1  | 91.5  | 91.4  | 84.4  | 73.6  | 64.8  | 57.4  |  |  |
|                                                                        |             | MCWB   | 46.3  | 47.4  | 51.9  | 55.8  | 62.8  | 68.0  | 67.9  | 67.7  | 63.4  | 60.8  | 56.5  | 51.3  |  |  |
|                                                                        | 2%          | DB     | 49.6  | 53.3  | 60.5  | 68.1  | 75.6  | 84.2  | 87.9  | 87.9  | 80.5  | 70.0  | 61.0  | 53.7  |  |  |
|                                                                        |             | MCWB   | 44.7  | 45.6  | 48.5  | 54.4  | 61.4  | 66.8  | 66.8  | 66.9  | 63.3  | 59.4  | 53.9  | 48.6  |  |  |
|                                                                        | 5%          | DB     | 46.9  | 50.2  | 57.2  | 64.5  | 73.1  | 81.0  | 84.6  | 84.8  | 77.0  | 67.2  | 58.6  | 51.3  |  |  |
|                                                                        |             | MCWB   | 43.5  | 44.0  | 47.3  | 52.9  | 60.4  | 65.6  | 66.4  | 66.1  | 62.4  | 58.0  | 52.5  | 47.3  |  |  |
|                                                                        | 10%         | DB     | 44.8  | 47.8  | 53.9  | 61.1  | 70.0  | 78.4  | 82.3  | 82.3  | 73.7  | 64.6  | 55.7  | 48.4  |  |  |
|                                                                        |             | MCWB   | 41.8  | 43.1  | 45.8  | 51.3  | 59.2  | 64.9  | 65.8  | 65.5  | 61.5  | 57.1  | 50.7  | 45.1  |  |  |
| (27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures | 0.4%        | WB     | 48.0  | 48.9  | 53.3  | 57.9  | 65.6  | 71.5  | 72.1  | 72.5  | 67.6  | 63.7  | 58.0  | 52.1  |  |  |
|                                                                        |             | MCDB   | 51.1  | 54.6  | 61.9  | 67.8  | 74.6  | 81.1  | 83.3  | 84.4  | 77.2  | 69.7  | 62.9  | 56.0  |  |  |
|                                                                        | 2%          | WB     | 45.8  | 46.6  | 50.4  | 55.7  | 63.4  | 69.1  | 70.2  | 70.0  | 65.4  | 61.1  | 55.3  | 49.7  |  |  |
|                                                                        |             | MCDB   | 48.3  | 51.1  | 57.3  | 65.4  | 72.7  | 79.7  | 82.1  | 82.0  | 75.8  | 67.4  | 59.8  | 52.9  |  |  |
|                                                                        | 5%          | WB     | 43.9  | 45.1  | 48.5  | 54.0  | 61.7  | 67.3  | 68.6  | 68.4  | 63.8  | 59.4  | 53.2  | 47.8  |  |  |
|                                                                        |             | MCDB   | 46.6  | 49.1  | 55.2  | 62.5  | 70.8  | 78.5  | 80.7  | 80.6  | 73.8  | 65.3  | 57.4  | 50.8  |  |  |
|                                                                        | 10%         | WB     | 41.9  | 43.6  | 46.9  | 52.3  | 60.0  | 65.5  | 67.3  | 66.9  | 62.5  | 57.8  | 51.2  | 45.7  |  |  |
|                                                                        |             | MCDB   | 44.4  | 47.2  | 52.8  | 59.7  | 68.5  | 76.4  | 79.4  | 79.2  | 71.9  | 63.4  | 55.3  | 48.4  |  |  |
| (35) Mean Daily Temperature Range                                      | 5% DB       | MDBR   | 14.1  | 15.6  | 17.8  | 19.3  | 21.7  | 23.9  | 26.0  | 26.2  | 21.9  | 19.2  | 16.2  | 14.0  |  |  |
|                                                                        |             | MCDBR  | 16.1  | 18.5  | 24.0  | 24.8  | 27.3  | 27.7  | 29.8  | 29.9  | 26.9  | 22.8  | 18.9  | 16.2  |  |  |
|                                                                        |             | MCWBR  | 11.7  | 12.1  | 14.1  | 13.5  | 14.7  | 13.6  | 14.1  | 14.0  | 13.6  | 13.8  | 12.4  | 11.9  |  |  |
|                                                                        | 5% WB       | MCDBR  | 14.7  | 16.6  | 21.9  | 23.1  | 25.4  | 26.4  | 27.6  | 27.9  | 24.6  | 20.9  | 17.1  | 14.8  |  |  |
|                                                                        |             | MCWBR  | 11.1  | 11.5  | 13.3  | 13.2  | 14.5  | 13.8  | 14.0  | 14.1  | 13.3  | 13.6  | 11.8  | 11.4  |  |  |
| (40) Clear-Sky Solar Irradiance                                        | taub        |        | 0.306 | 0.331 | 0.376 | 0.426 | 0.422 | 0.399 | 0.405 | 0.418 | 0.402 | 0.367 | 0.333 | 0.307 |  |  |
|                                                                        | taud        |        | 2.432 | 2.353 | 2.265 | 2.132 | 2.206 | 2.311 | 2.291 | 2.251 | 2.283 | 2.370 | 2.432 | 2.441 |  |  |
|                                                                        | Ebn at Noon |        | 269   | 278   | 275   | 268   | 272   | 278   | 275   | 268   | 264   | 260   | 255   | 258   |  |  |
|                                                                        | Edh at Noon |        | 25    | 31    | 38    | 47    | 45    | 41    | 41    | 42    | 37    | 31    | 25    | 23    |  |  |
| (44) All-Sky Solar Radiation                                           | RadAvg      |        | 559   | 808   | 1126  | 1457  | 1776  | 2075  | 2172  | 1925  | 1407  | 996   | 645   | 489   |  |  |
|                                                                        | RadStd      |        | 79    | 107   | 152   | 166   | 141   | 155   | 122   | 127   | 123   | 107   | 65    | 89    |  |  |

## Historical Trends

|                         | DBAvg | Heating |        | Cooling |       |       | Degree-Days |       |       |       |
|-------------------------|-------|---------|--------|---------|-------|-------|-------------|-------|-------|-------|
|                         |       | 99% DB  | 99% DP | 1% DB   | 1% WB | 1% DP | HDD50       | HDD65 | CDD50 | CDD65 |
|                         |       | (d)     | (e)    | (f)     | (g)   | (h)   | (i)         | (j)   | (k)   | (l)   |
| Station Only            | N/A   | N/A     | -2.03  | N/A     | N/A   | N/A   | N/A         | N/A   | N/A   | N/A   |
| Regional (21 neighbors) | +0.90 | N/A     | N/A    | N/A     | N/A   | N/A   | N/A         | -221  | +184  | N/A   |

Nomenclature: See separate page